Smart Library Request Workflow in ServiceNow

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1.INTRODUCTION:

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OVERVIEW OF THE PROJECT:

- The Smart Library Request Workflow in ServiceNow project focuses on improving the efficiency of library management in educational institutions.
- In many libraries, book requests and approvals are handled manually, which can lead to delays and errors. Students often find it difficult to know the availability of books, and librarians face challenges in tracking borrow requests.
- This project provides a digital solution where students can request books through the system. Librarians can review and approve or reject the requests using an automated workflow.
- This system improves transparency, reduces manual effort, and makes library management more efficient.

2. IDENTIFICATION PHASE:

INTRODUCTION:

- Many educational institutions still manage library activities manually. Students need to check book availability by directly visiting the library, and librarians must manually track borrow and return records.
- This manual process can lead to delays, lack of transparency, and difficulty in maintaining accurate records.

PROBLEM IDENTIFICATION:

- Students often face problems in knowing which books are available. Librarians also find it difficult to track borrow requests, approve them, and update the book status manually.
- This process consumes time and may result in errors.

OBJECTIVES OF THE PROJECT:

- To create a digital library request system.
- To allow students to request books online.
- To enable librarians to approve or reject requests.
- To automate the process of updating book status.

3. REQUIREMENTS ANALYSIS:

- Requirement analysis is an important stage where the needs and requirements of the system are identified before development begins.

Functional Requirements:

- Students should be able to request books through the system.
- Librarians should be able to manage book records.
- Librarians should approve or reject borrow requests.
- Book status should update automatically when issued or returned.
- Reports should show the most borrowed books.

Non-Functional Requirements:

- The system should be user friendly.
- Data should be secure and reliable.
- The system should provide quick response time.

Software Requirements:

- ServiceNow platform
- Google Sheets or Microsoft Excel
- Web browser such as Chrome or Edge

Hardware Requirements:

- Computer or laptop
- Internet connection
- ServiceNow instance access

4. PROJECT PLANNING PHASE:

- The Project planning defines how the project will be executed.

Project Schedule:

- The project is divided into several stages such as:

❖ Problem identification

❖ Requirement analysis

❖ System design

❖ Development and implementation

❖ Testing

❖ Documentation

❖ Final demonstration

Planning Activities:

- Creating roles for students and librarians
- Designing tables for storing book and request data
- Creating workflows for approval process
- Configuring security rules
- Creating reports and testing the system

Risk Management:

- Some possible risks during project implementation include:
  - Incorrect field mapping during transform map creation
  - Data format errors in the import file
  - Configuration mistakes in ServiceNow tables

Resources Required:

- ServiceNow Personal Developer Instance
- Sample library data
- Documentation tools

5. PROJECT DEVELOPMENT PHASE:

- ✓ The development phase focuses on implementing the actual solution in the ServiceNow platform.

Step 1: Creating Roles

- Two roles are created:
  - ✓ Student role – used for requesting books
  - ✓ Librarian role – used for managing books and approvals

Step 2: Creating Tables

- Two custom tables are created:
  - ✓ Book Table – stores book details
  - ✓ Borrow Request Table – stores book request information

Step 3: Adding Fields

Fields are added to the tables.

- Book Table Fields
  - ✓ Title
  - ✓ Author
  - ✓ ISBN
  - ✓ Status

- Borrow Request Table Fields:
  - ✓ Requested
  - ✓ Book
  - ✓ Request Date
  - ✓ Status

Step 4: Configuring Reference Qualifier

- A reference qualifier is added so that students can only select books that are available.

Step 5: Implementing Workflow

- Using Flow Designer, a workflow is created where:
  - ➢ Students submit borrow requests
  - ➢ Librarians approve or reject requests
  - ➢ If approved, the book status is automatically changed to Issued

6. PROJECT DESIGN PHASE:

- This phase describes the system structure and how components interact

System Design Overview:

- The system is designed to automate library operations using ServiceNow features.

Key Design Components:

✓ **Roles**

Roles define the permissions for students and librarians.

✓ **Tables**

Tables store book information and borrow request records.

✓ **Flow Designer**

Used to automate the approval process and update book status

✓ **UI Policies**

Used to control form behavior and improve usability.

✓ **Access Control Rules**

Used to provide security and restrict access based on roles.

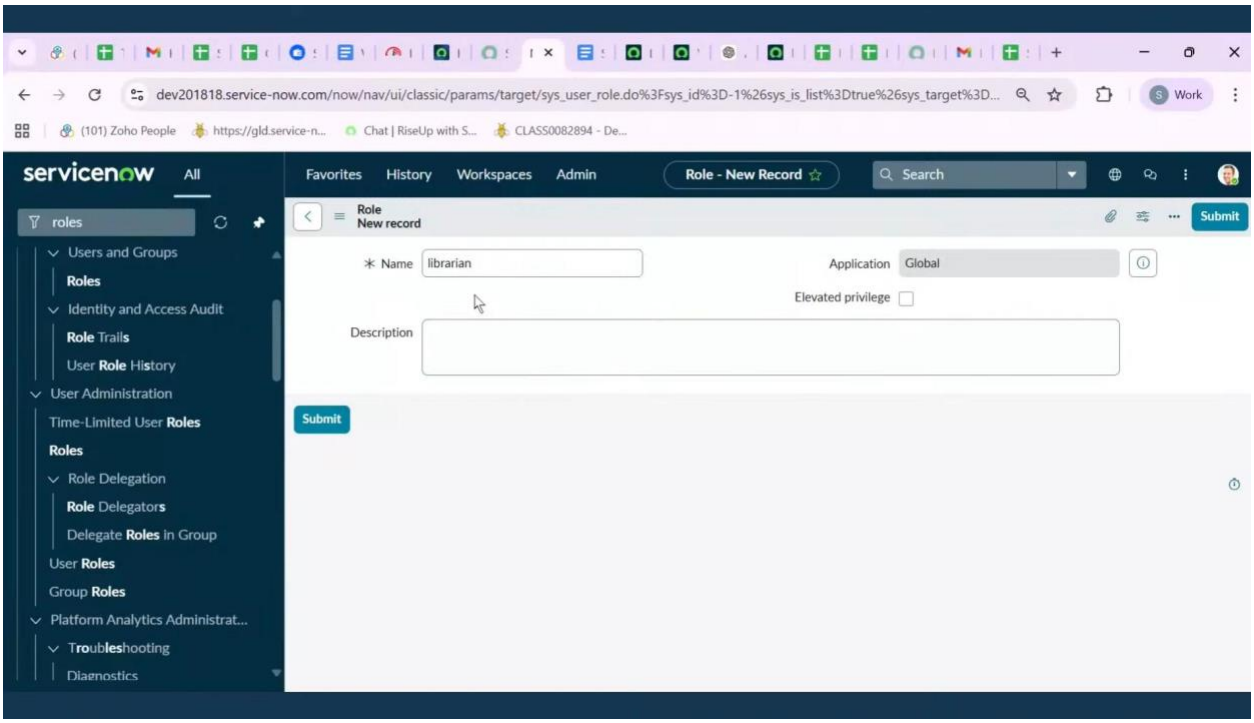
✓ System Workflow

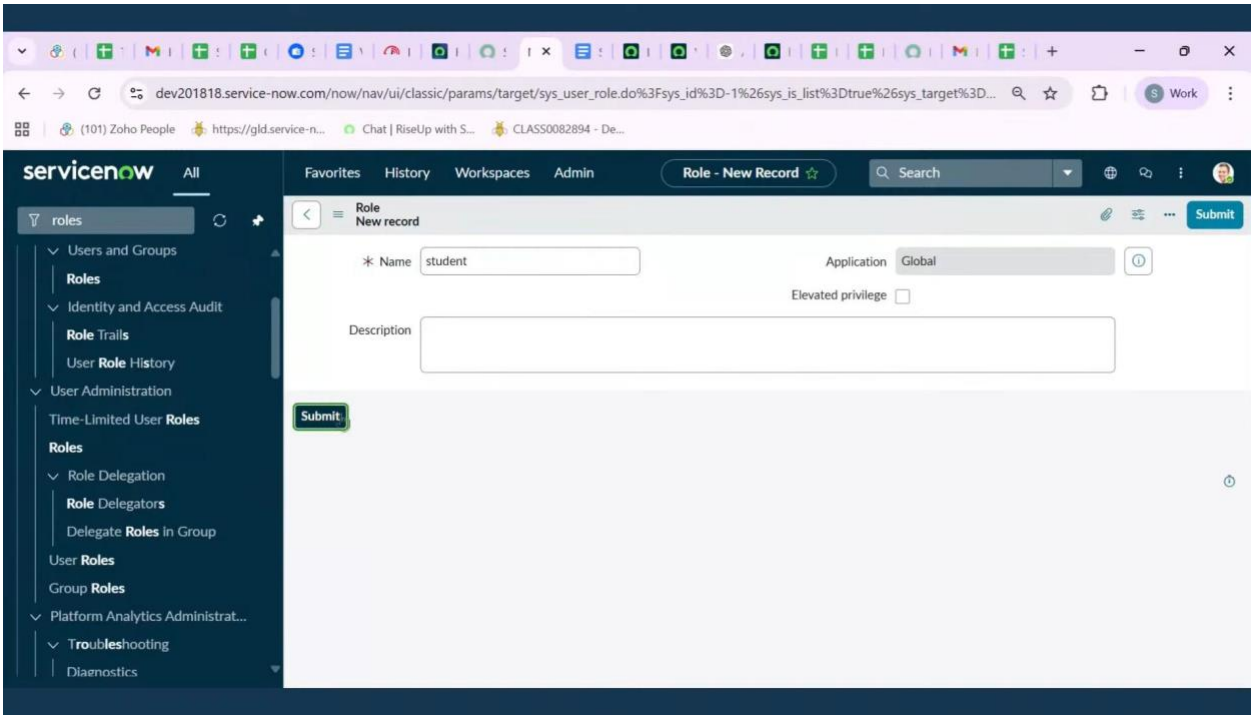
- ❖ Student submits borrow request.
- ❖ Request is sent to librarian.
- ❖ Librarian approves or rejects the request.
- ❖ Book status changes automatically.
- ❖ Reports show book usage statistics.

7. PROJECT DOCUMENTATION:

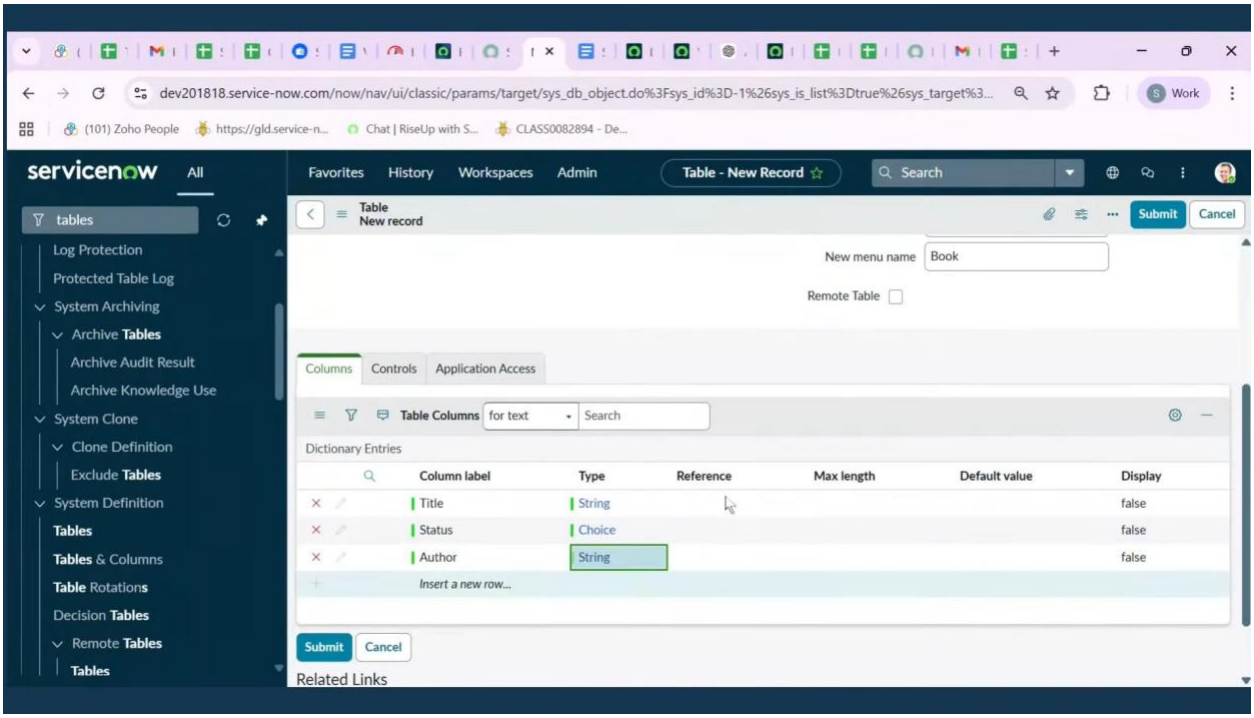
Project documentation is an essential part of the project as it records all the information related to the project development process

Step 1: Create Roles

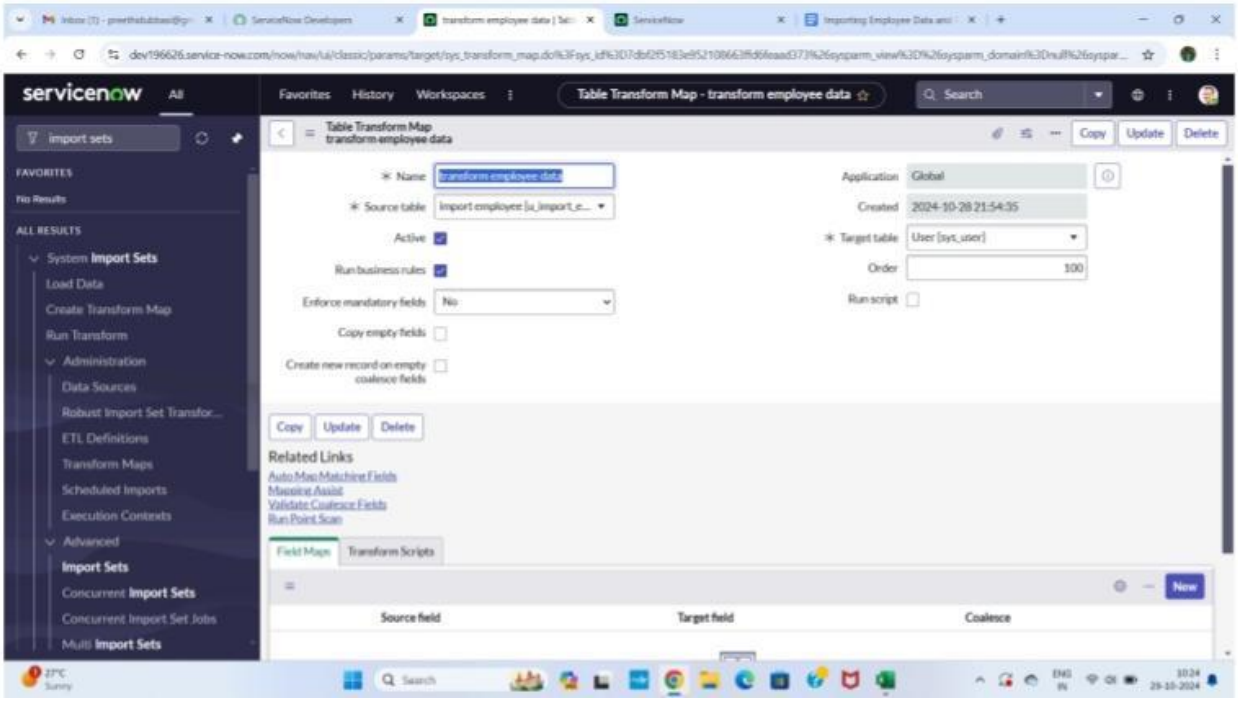


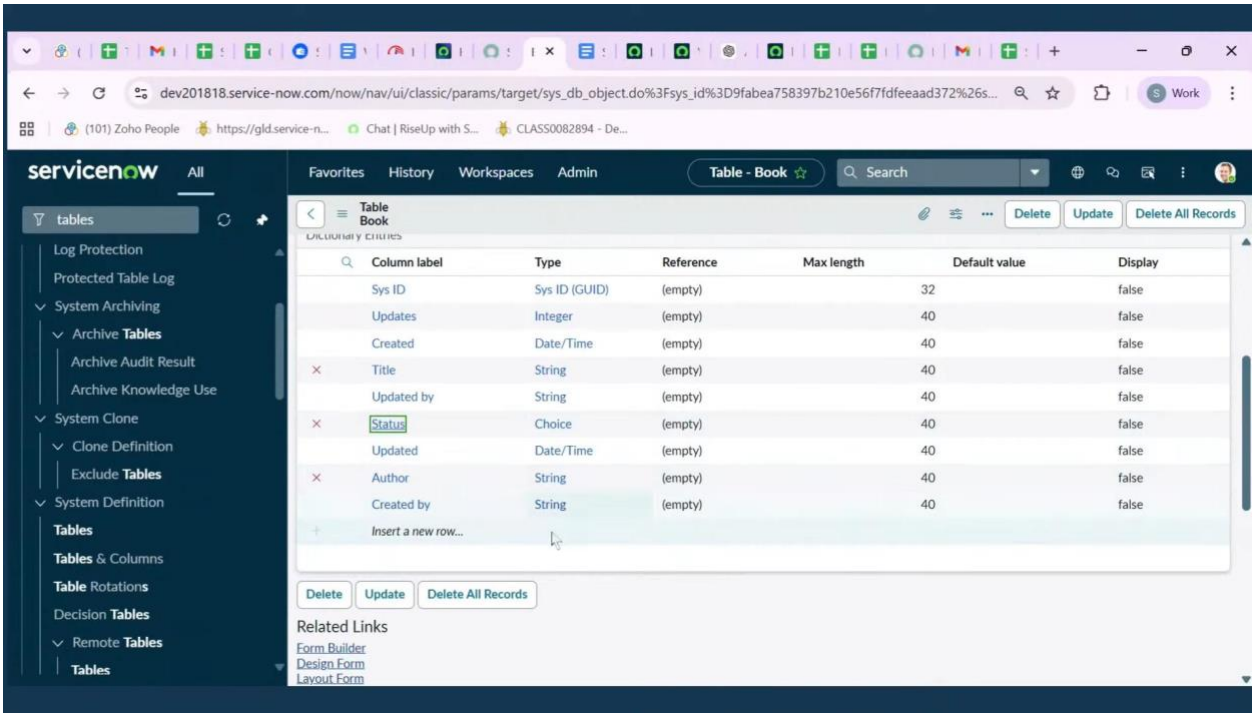


Step 2: Add Fields

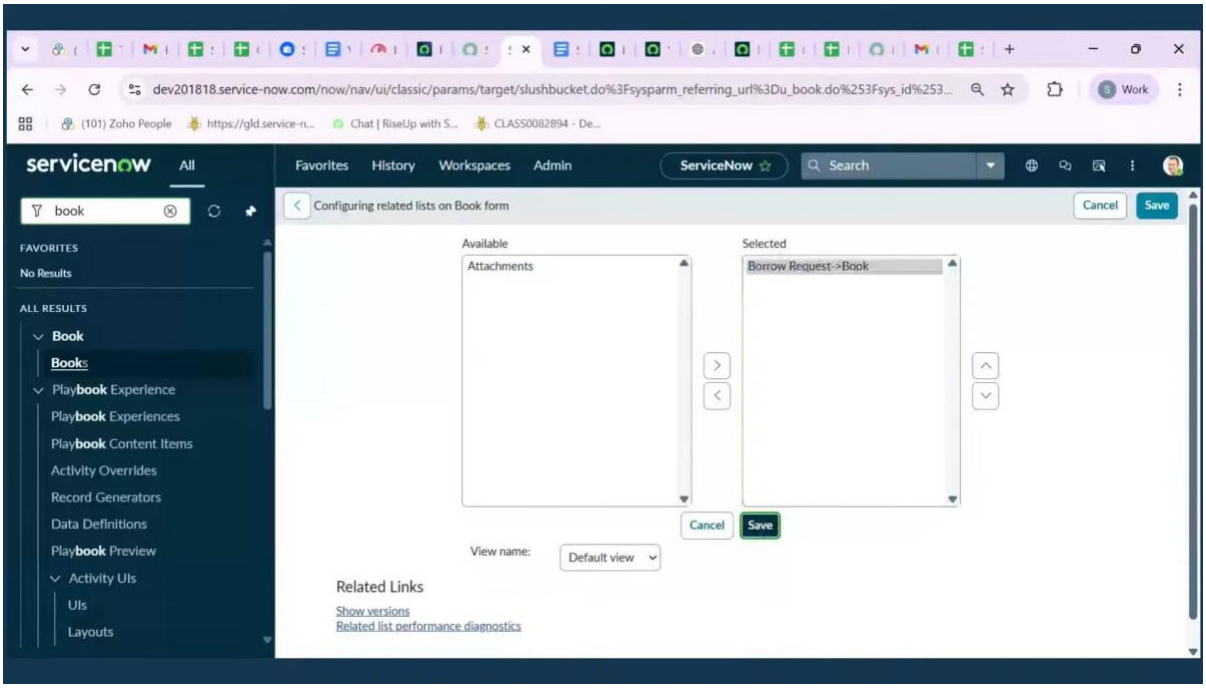


Step 3: Status Updated

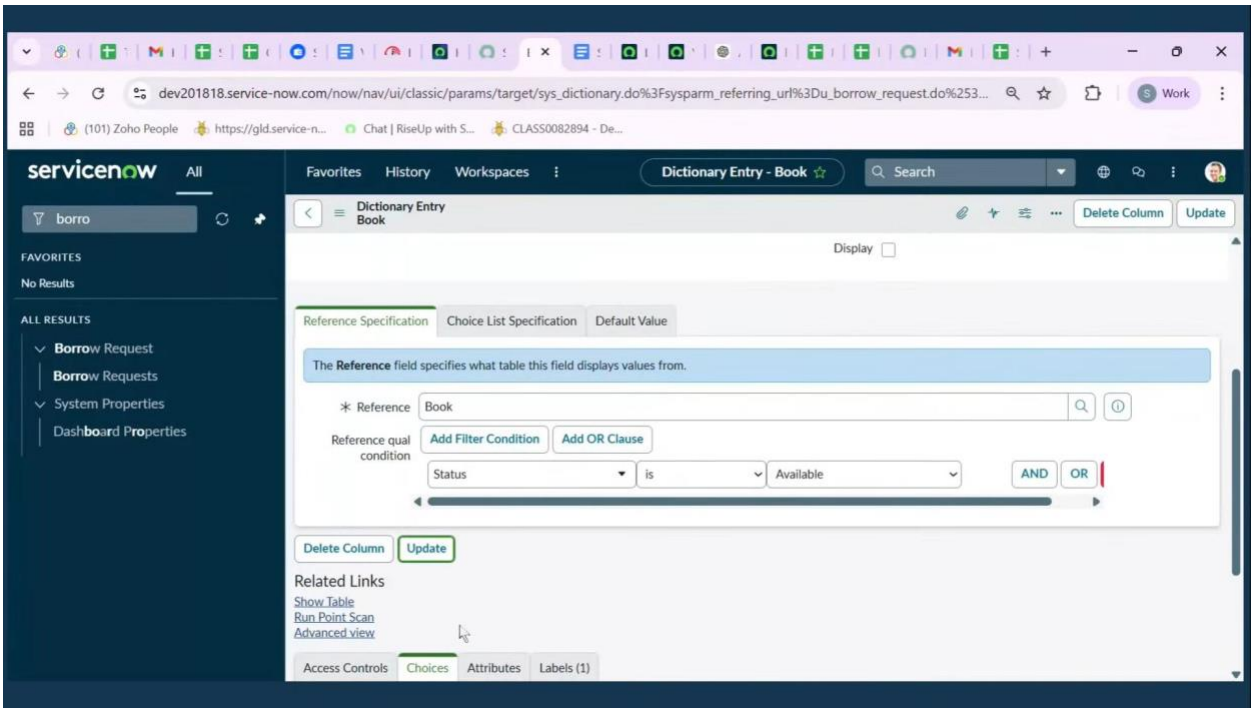




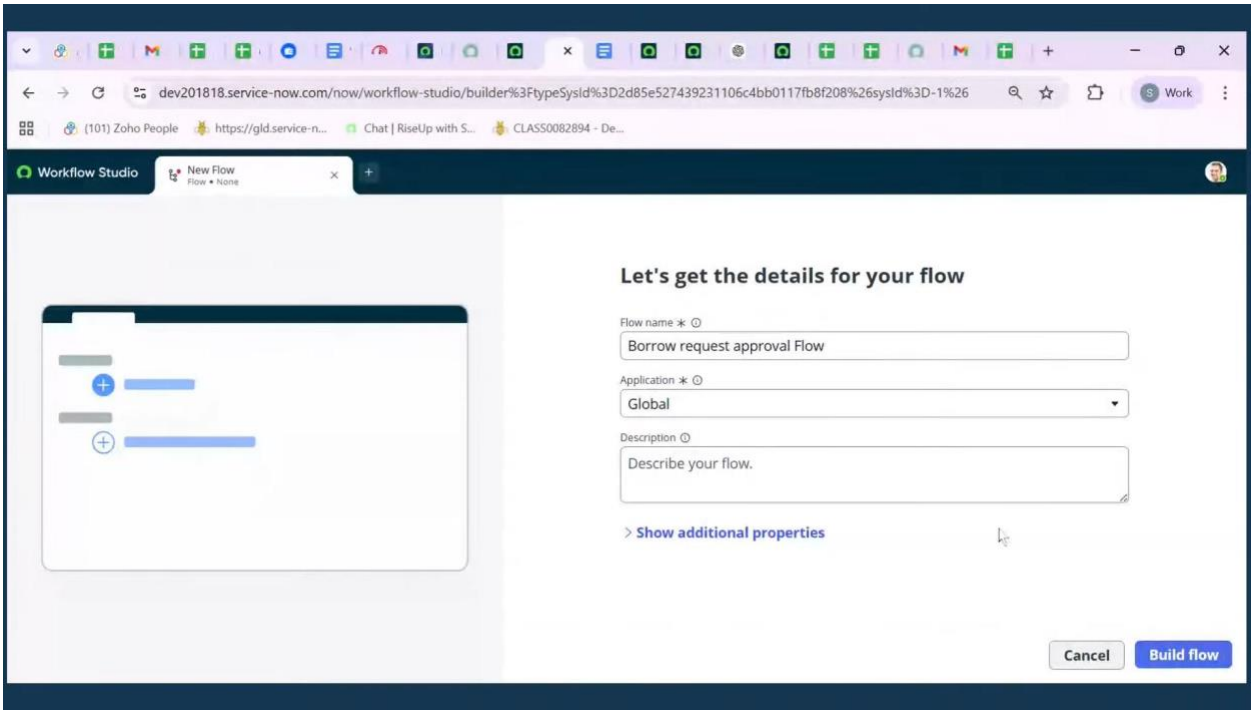
Step 4: Configure Refrence Qualifier



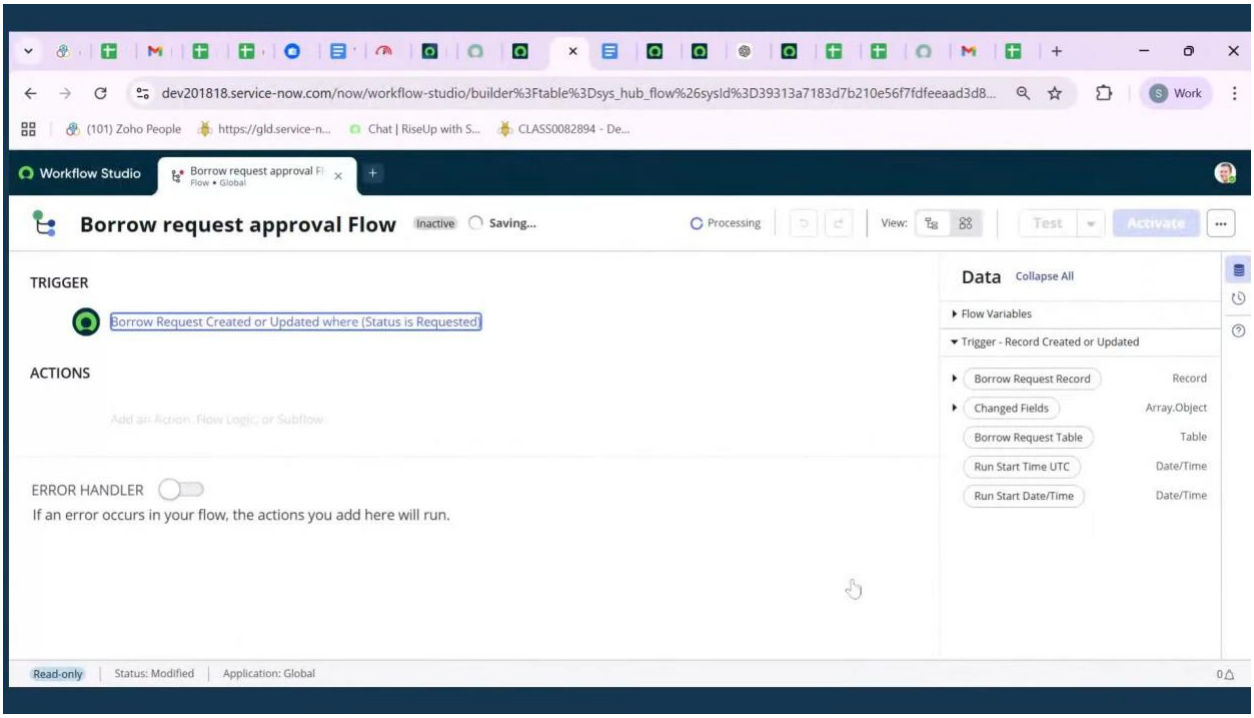
Step 5: Update Dictionary Book



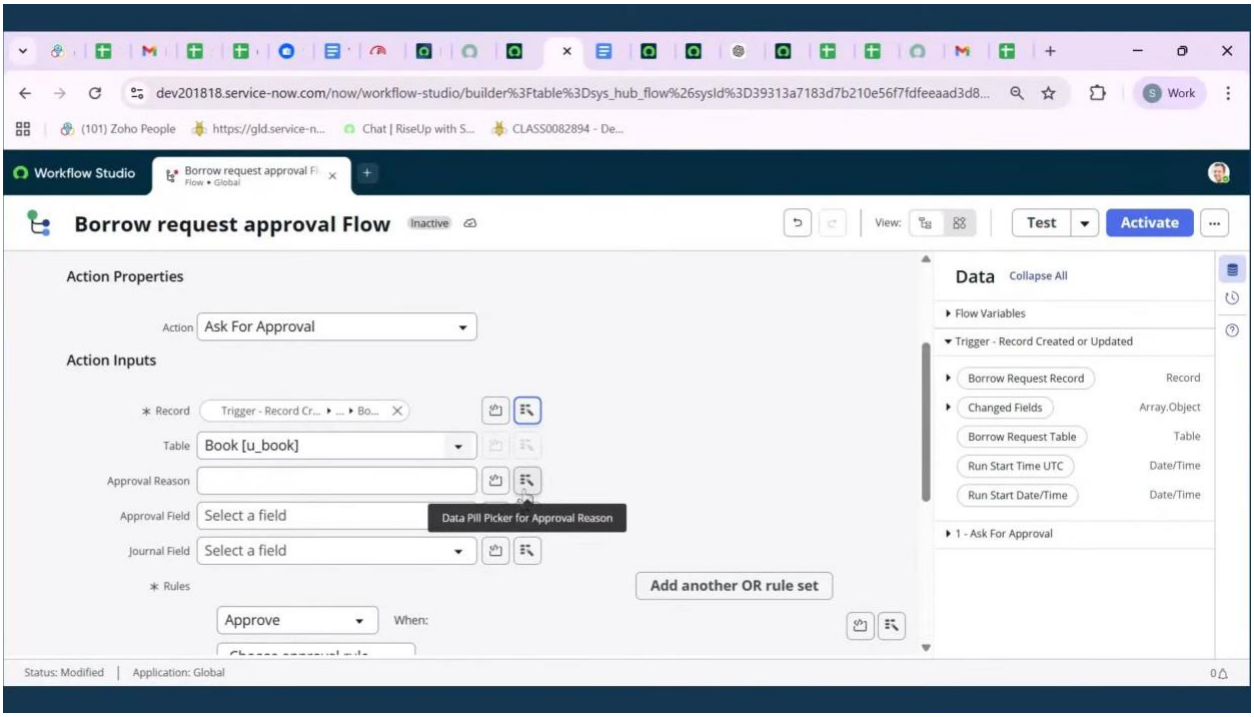
Step 6: Flow Designer



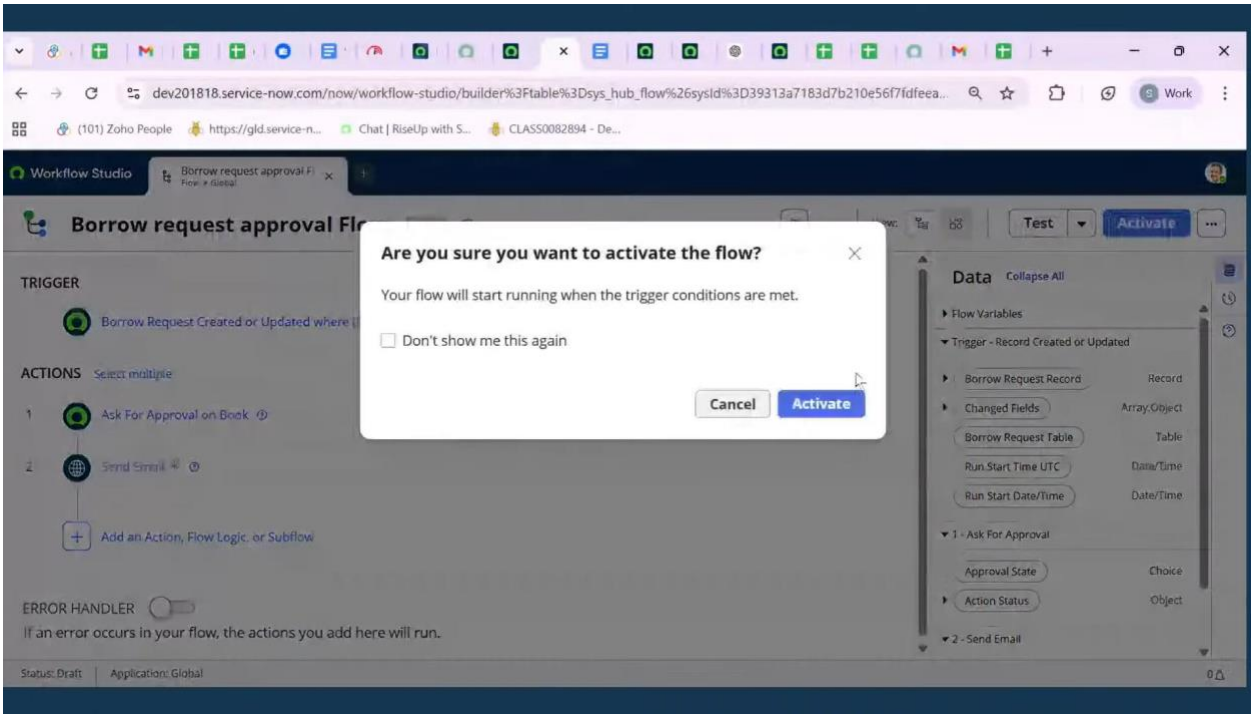
Step 7: Add a Trigger



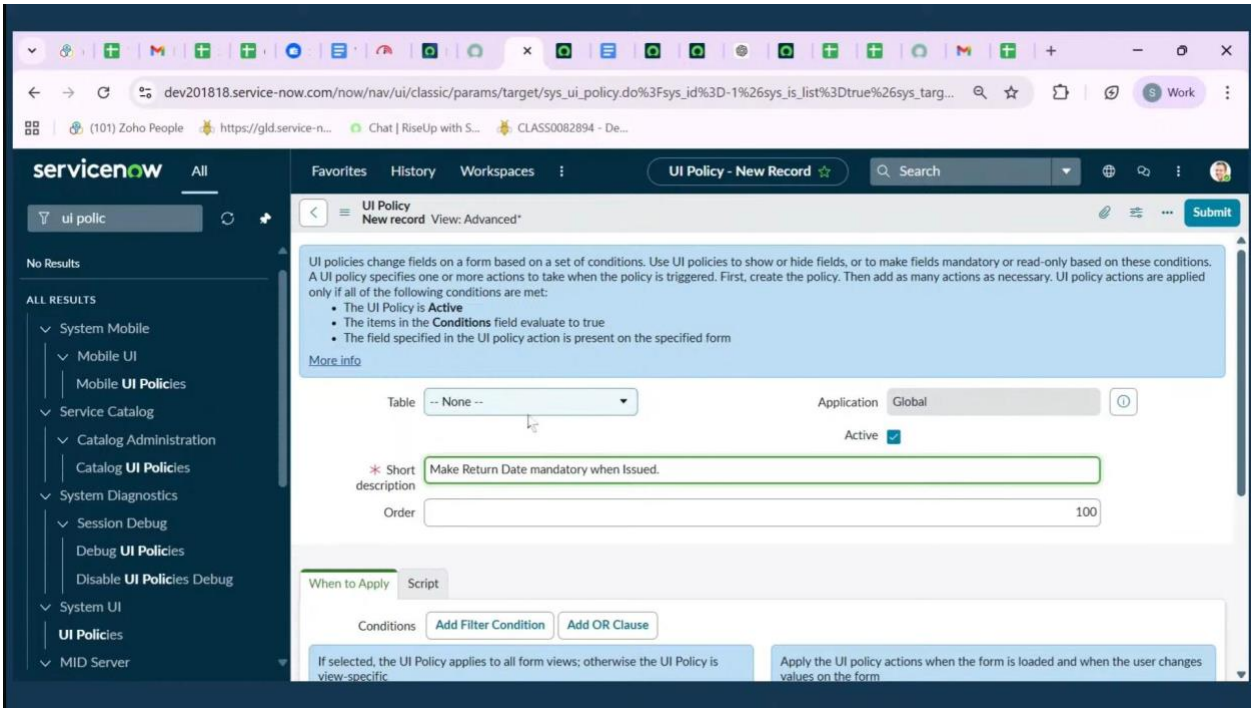
Step 8: Borrow Request Record



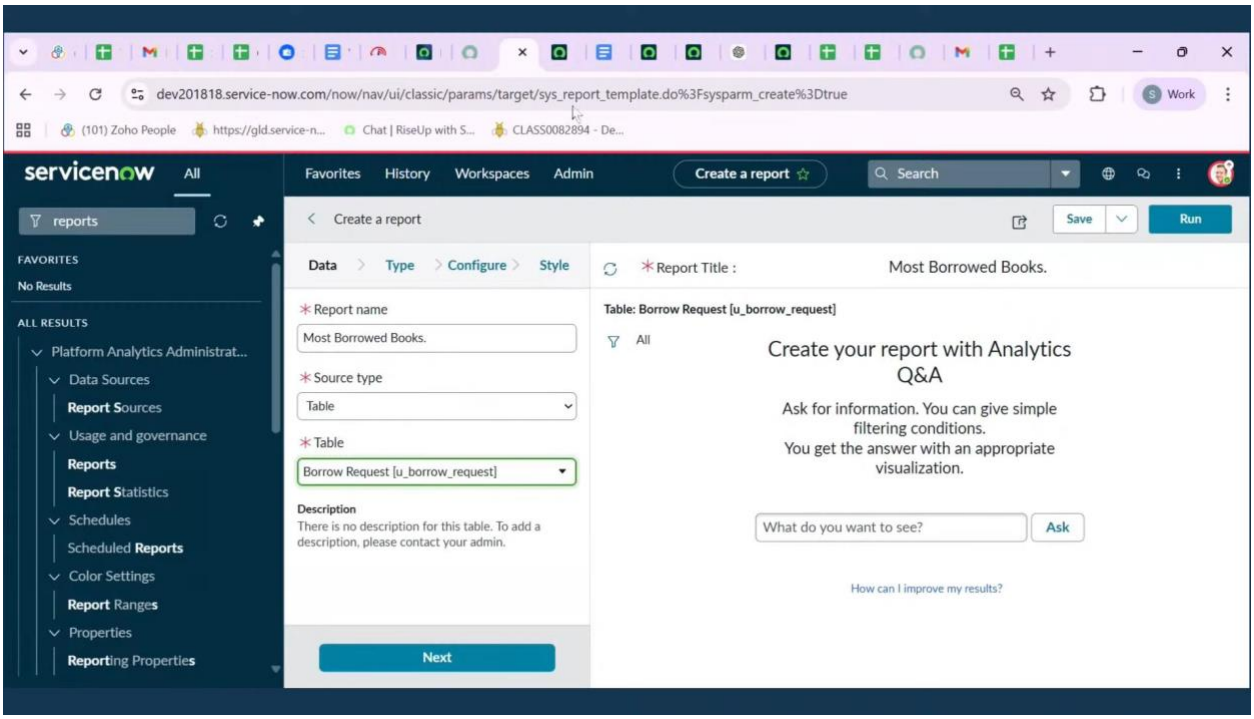
Step 9: Activate the Action

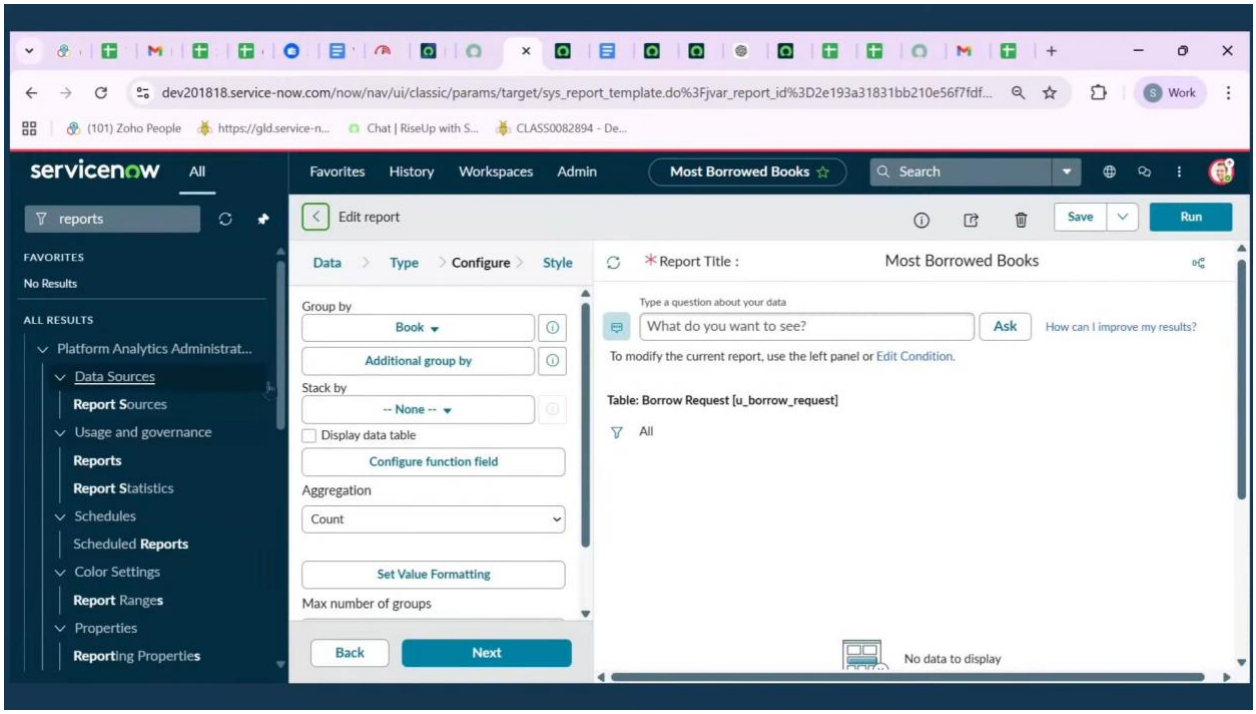


Step 10: UI Policy



Step 11: Create Report

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Search
reports
FAVORITES
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ALL RESULTS
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Data Sources
Report Sources
Usage and governance
Reports
Report Statistics
Schedules
Scheduled Reports
Color Settings
Report Ranges
Properties
Reporting Properties
Edit report
Save
Run
Data
Type
Configure
Style
Report Title :
Most Borrowed Books
Group by
Book
Additional group by
Stack by
-- None --
Display data table
Configure function field
Aggregation
Count
Set Value Formatting
Max number of groups
Back
Next
Type a question about your data
What do you want to see?
Ask
How can I improve my results?
To modify the current report, use the left panel or Edit Condition.
Table: Borrow Request [u_borrow_request]
All
No data to display

8. PROJECT DEMONSTRATION:

The project demonstration phase is where the completed system is presented and explained in Video Format

- Show the Book table and available books.
- Login as a Student and submit a borrow request.
- Show the request status as Requested.
- Login as a Librarian and approve the request.
- Show that the book status changes to Issued.
- Display the Most Borrowed Books report

FINAL RESULT:

- The system successfully automates the library book request process, improves efficiency, and provides better tracking of library resources

9. CONCLUSION:

- The Smart Library Request Workflow in ServiceNow simplifies the library management process by automating book requests, approvals, and tracking.
- This improves efficiency for both students and librarians.